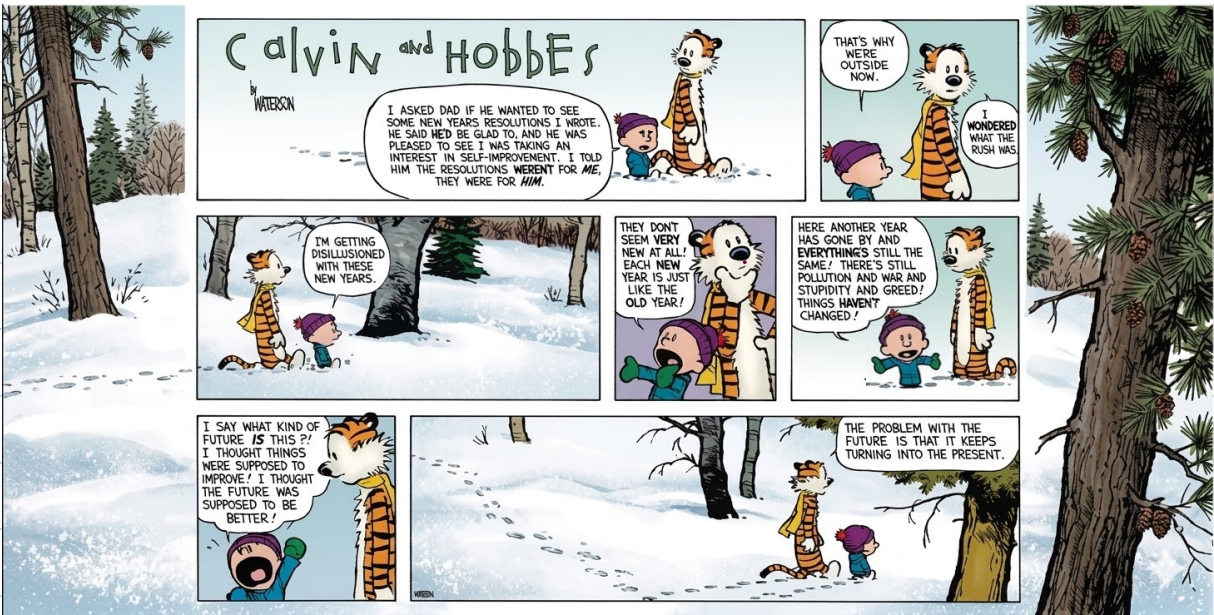




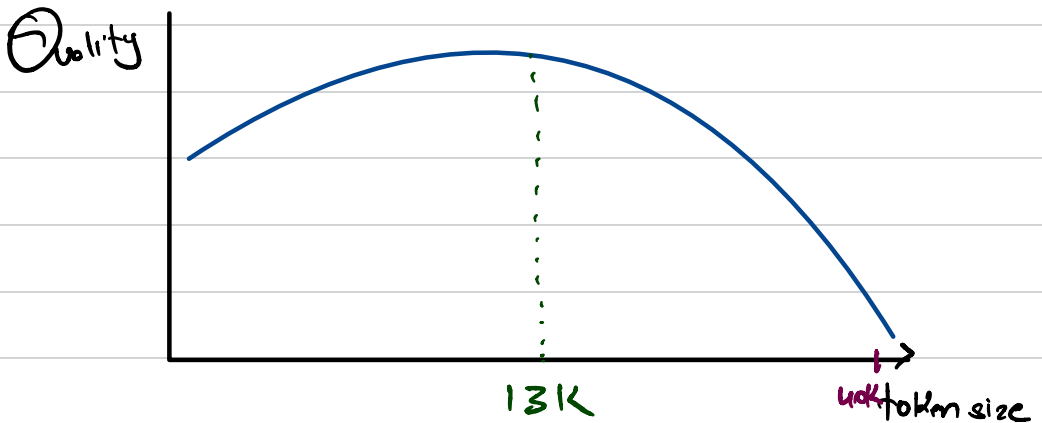
# Homework June 02, 2026 -> June 04, 2026 (hopefully)

1. Design the tool-schema for Weather API
2. Function for talking to the API
3. Design the flow for your BE to talk to LLM, and send response of API back to LLM to get the final answer.

## Models

[View all →](#)

| Name                                   | Size / Usage | Context | Input |
|----------------------------------------|--------------|---------|-------|
| qwen3-embedding:latest                 | 4.7GB        | 40K     | Text  |
| qwen3-embedding:0.6b                   | 639MB        | 32K     | Text  |
| qwen3-embedding:4b                     | 2.5GB        | 40K     | Text  |
| qwen3-embedding:8b <span>latest</span> | 4.7GB        | 40K     | Text  |



Test - data → Golden Set

$$DA \int \frac{SA}{1K}, \frac{SB}{1K} \rightarrow \text{Sim Score}$$

$$DB \int \frac{SA}{10K}, \frac{SB}{10K} \rightarrow \text{Sim Score}$$

DB  $\int \frac{SA}{20K}, \frac{SB}{20K} \rightarrow \text{Sim Score}$

A  
  
 10K  
 1K item

  
 B



C 768

Steps to follow for RAG + LLM:

1. Load/Download Data.
2. Chunk-data, embedding-gemma for embedding model.
3. ChromaDB (offline + online - chroma cloud)
4. Chroma DB - Create a collection
5. Add all the indexed vectors to ChromaDB
6. Query -> Embedding model -> embedding.
7. Retrieve text from all these chunks -> LLM

Vector DB  $\rightarrow$  Numpy array

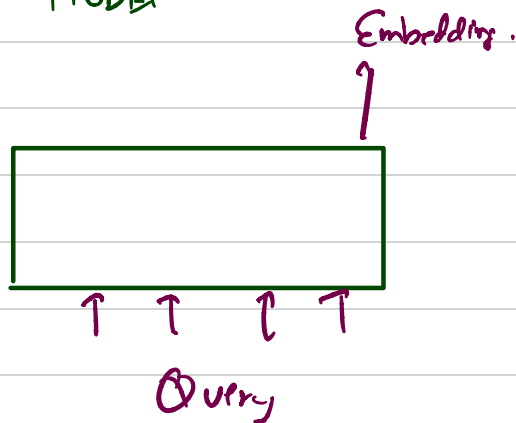
50K, 768  
 HNSW, FAISS  $\rightarrow$  Process the vectors

np.save

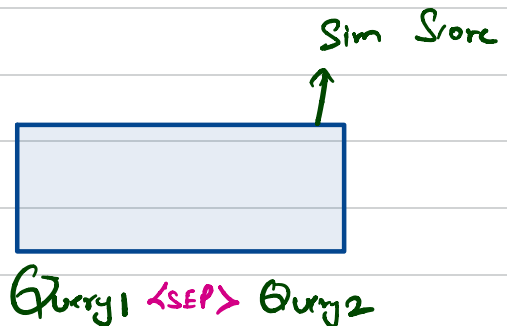
Chunking Strategies -> Recursive



## EMBEDDING MODEL



## CROSS-ENCODER

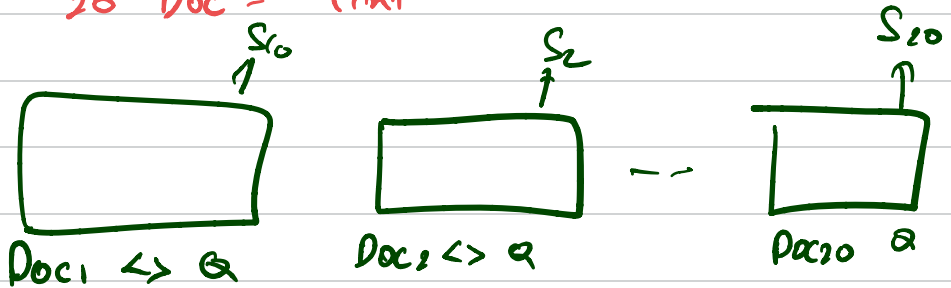


SOLK

- Query  $\rightarrow$  20

① - Cosine Sim - Top 20 most relevant documents

② Query - text  
20 Doc - text



Follow up Query: Why was that function not optimal?

### 3 Days. IIT Roorkee. AI Agents. Hackathon. Networking. Memories.

Imagine stepping onto the iconic IIT Roorkee campus and spending an entire weekend building, learning, collaborating, and competing alongside fellow learners from across the country.

#### What awaits you?

##### Day 0 – Arrival & Networking (26 June)

- Check in to your IIT Roorkee accommodation
- Program orientation and campus briefing
- Dinner and networking with fellow learners before the action begins

##### Day 1 – Learn, Build & Explore (27 June)

- Inaugural ceremony with IIT Roorkee and Scaler
- Deep dive into the foundations of AI Agent Building
- Hands-on workshops where you'll build agents under expert guidance
- Guided IIT Roorkee campus tour, including research labs and key departments
- Hackathon kickoff: team formation, problem statement reveal, and strategy discussions

##### Day 2 – Hackathon & Showcase (28 June)

- Full-scale hackathon where you'll bring your AI agent to life
- Team collaboration, brainstorming, and final sprint sessions
- Present your solution before judges and peers
- Certification ceremony and closing session
- Walk away with not just a certificate, but an unforgettable IIT Roorkee experience

 This isn't a typical classroom event.

You'll get to:

-  Experience life on the IIT Roorkee campus
-  Build real AI agents hands-on
-  Participate in a live hackathon
-  Network with ambitious peers and industry professionals
-  Learn directly through immersive, practical sessions
-  Create memories that extend beyond the classroom

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 Seats are limited, and we would love to see you there!

